🏆 AI League #3: MCP Servers for Indian Financial Intelligence

Build a Production-Grade MCP Server with Real Auth & Cross-Source Reasoning

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📌 Background

An Indian retail investor or analyst evaluating a stock today must manually check NSE/BSE for live prices, visit Moneycontrol or Screener for fundamentals, open AMFI for mutual fund NAVs, read SEBI/BSE filings for corporate announcements, track RBI data for macro indicators, and scan financial news portals for sentiment. Each source has a different interface, different data format, and no interoperability.

The good news: much of this data is freely available through public APIs and open data sources. The bad news: there is no unified way for an AI assistant to discover and use all of these capabilities together.

The **Model Context Protocol (MCP)** solves this. An MCP server wraps these existing APIs into structured tools, resources, and prompts that any MCP-compatible client can discover and use. But financial data is sensitive and role-dependent: a free-tier user shouldn't consume premium API quotas, and a retail investor shouldn't see the same depth as a research analyst. This makes authentication and authorisation essential engineering concerns.

🎯 The Challenge

Build a **production-grade MCP server** that wraps existing free Indian financial APIs into a unified intelligence layer. The server exposes tools for NSE/BSE market data, fundamental analysis, mutual fund data, news sentiment, regulatory filings, and macro-economic indicators — all as MCP primitives (tools, resources, prompts).

The server must implement **OAuth 2.1-based authentication** and **role-based authorisation** per the MCP authorisation specification, so different user tiers see different capabilities and rate limits.

Key constraint: Spend most effort on MCP server design, auth implementation, and cross-source reasoning — not on building financial models from scratch. The underlying data comes from existing free APIs. Your job is to wrap them intelligently, add cross-source analysis, and secure the whole thing properly.

⚡ **MVP Mindset:** This is a hackathon, not a production deployment. Focus your time on the MCP server design, auth flow, and cross-source reasoning — the three things that get judged hardest. For peripheral tasks (PDF parsing, analyst estimates, data ingestion), take shortcuts if time consuming: hardcode sample data, use LLM extraction, mock what you can't fetch. *A working demo with smart architecture beats a half-finished system that tries to do everything from scratch!*

🔀 Choose Your Use Case

All teams build the **same base MCP server layer** (APIs, auth, tiered access, caching). What differs is the **use case** you solve on top — this determines which tools you prioritize, what cross-source reasoning looks like, and where your engineering depth goes.

Pick ONE of three use cases below. Each forces different architectural decisions. Choose based on what excites you and where you want to push yourself.

- Shared Base Layer (Required for ALL Use Cases)
1. [PS1] USE CASE 1: Financial Research Copilot
2. [PS2] USE CASE 2: Portfolio Risk & Alert Monitor
3. [PS3] USE CASE 3: Earnings Season Command Center

📦 Shared Base Layer (Required for ALL Use Cases)

This is what every team must build regardless of which use case they pick.

Pre-Built APIs to Integrate (Use at Least 4)

The following free APIs provide the raw data layer. Teams must integrate at least four. The challenge is not calling these APIs — it's exposing them as well-designed MCP tools with proper auth, error handling, caching, and cross-source reasoning on top.

API / Source	What It Provides	Free Tier	Modality
NSE India (stock-nse-india npm / direct)	Live quotes, indices, F&O data, historical OHLCV, option chains, top gainers/losers	Free, no key (rate limited)	JSON, Tabular

yfinance (.NS / .BO tickers)	Historical prices, dividends, splits, financials, balance sheet, cash flow for NSE/BSE stocks	Free, no key (unofficial)	Tabular, JSON
MFapi.in	Mutual fund NAV history, scheme search, fund house data for all AMFI-registered schemes	Free, no key	JSON
Alpha Vantage (India tickers: .BSE)	Stock prices, technicals (SMA, RSI, MACD), fundamentals, news + sentiment	25 req/day (free key)	JSON, Tabular
Finnhub (India support)	Company news, earnings, recommendations, insider transactions	60 req/min (free key)	JSON, Text
BSE India (bseindia.com APIs)	Corporate announcements, board meetings, results, annual reports (PDFs)	Free, no key	PDF, JSON
RBI DBIE (data.rbi.org.in)	Macro data: repo rate, CPI inflation, GDP, forex reserves, money supply	Free (downloadable datasets)	CSV, JSON
data.gov.in	Government open data: economic indicators, sector-wise statistics	Free (API key required)	JSON, CSV
NewsAPI.org / GNews	Indian financial news articles with search and filtering	100 req/day (free key)	Text, Images

*Teams may substitute equivalent APIs (e.g., Twelve Data for technicals, Marketaux for sentiment). The requirement is at least **4 distinct data sources** covering at least **3 data types** (market data/tabular, text/news, and documents/PDFs or macro data).*

Core Requirement: Authentication & Authorisation

<u>This is non-negotiable and common across all use cases.</u>

Authentication Requirements:

- OAuth 2.1 with PKCE — Mandatory for all MCP clients (public clients)
- Protected Resource Metadata (RFC 9728) — Serve `/.well-known/oauth-protected-resource` advertising the authorisation server and available scopes

- Bearer Token Validation — All operations validate token signature, expiry, audience (RFC 8707), and scopes
- 401 with Discovery — Unauthenticated requests return 401 with `WWW-Authenticate` header containing `resource_metadata` URL

Authorisation: Tiered Access Control

The server must enforce at least three user tiers:

	Free User	Premium User	Analyst
Market Data	Quotes, price history, indices, top movers	All Free + technicals, option chains	All Premium + everything
Fundamentals	❌	Financials, ratios, shareholding, quarterly results, corporate actions	All
Mutual Funds	Search + NAV lookup	All Free + comparison, overlap	All
News	Company news, market news	All Free + sentiment analysis	All
Filings	❌	❌	Full filing retrieval (PDFs)
Macro	❌	Current snapshot (RBI rates, inflation)	Full historical time series
Cross-Source Tools	❌	❌	All cross-source reasoning tools
Rate Limit	30 calls/hour	150 calls/hour	500 calls/hour

Insufficient permissions → HTTP 403 with `insufficient_scope`. Rate limit exceeded → HTTP 429 with `Retry-After` header.

Clarification on cross-source tools vs multi-API tools: *"Cross-source reasoning tools" are the ones explicitly labeled as such in each use case (e.g., `cross_reference_signals`, `portfolio_risk_report`, `earnings_verdict`). Other tools that internally call multiple APIs to serve their function (e.g., `check_mf_overlap` calls MF + holdings data) are categorized by their primary scope, not as cross-source.*

Cross-source tools always require an Analyst tier. See per-use-case tier addendum tables for specifics.

Scope Design:

- `market:read` — Live quotes, price history, indices, movers
- `fundamentals:read` — Financial statements, ratios, shareholding, results
- `technicals:read` — Technical indicators, option chains
- `mf:read` — Mutual fund NAV, search, comparison
- `news:read` — News articles and sentiment
- `filings:read` — List filings
- `filings:deep` — Retrieve full filing documents (PDFs)
- `macro:read` — RBI rates, inflation, forex (current)
- `macro:historical` — Full historical macro time series
- `research:generate` — Cross-source reasoning tools
- `watchlist:read` / `watchlist:write` — Personal watchlist
- `portfolio:read` / `portfolio:write` — Portfolio holdings, summaries, and risk tools (PS2)

Technical Requirements (All Use Cases)

MCP Protocol Compliance:

- Streamable HTTP transport for remote deployment
- Capability negotiation — tier-aware: Free users discover fewer tools than Analysts
- Pagination for news results, filing lists, price history
- Resource subscriptions (optional but encouraged) — notify clients when subscribed resources change (e.g., watchlist ticker news, portfolio alerts, new filings posted). Core to PS2; bonus for PS1/PS3.
- Structured error responses for API failures, rate limits, permission denials

Identity Provider Integration:

- OAuth 2.1-compliant auth server (Keycloak recommended; Auth0/Okta free tier acceptable)
- Auth server separate from MCP server (Resource Server pattern)
- No token pass-through — MCP server manages its own upstream API keys

API Key Management & Caching:

- MCP server holds API keys for upstream providers in its own config — never exposed to clients
- Rate limiting must respect both user tier limits AND upstream API quotas

- Aggressive caching: quotes cached ~60s, news ~30min, financials ~24h, filings permanently

Data Layer:

- Cache for upstream API responses (TTL-based per data type)
- User data store for watchlists and cached research (scoped per user)
- Audit log: every tool invocation logged with user identity, tier, tool name, timestamp

Constraints & Rules (All Use Cases)

- All tool outputs must cite their data source (e.g., "Source: NSE India", "Source: MFapi.in scheme 125497", "Source: BSE Filing #12345")
- Tools must return **structured JSON**, not free-form text. The AI client handles narrative; the server provides data
- Cross-source tools must explicitly list which sources confirmed or contradicted each finding
- The MCP server must hold all upstream API keys internally — never expose them to clients
- Responses must not constitute financial advice. Include appropriate disclaimers
- Handle upstream API downtime gracefully (return cached data or clear error, not crash)

🔀 [PS1] USE CASE 1: Financial Research Copilot

"The full-stack analyst's assistant"

***Pick this if** you want to build the broadest, most general-purpose financial intelligence tool. You'll flex on breadth of integration, synthesis across many data types, and creating an experience where an analyst can say "tell me everything about Reliance" and get a genuinely useful answer.*

What You Build

A general-purpose research copilot. The user connects an MCP client (Claude Desktop, VS Code), authenticates, and can run queries ranging from quick stock checks to full deep-dive research briefs that pull data from 5+ sources.

MCP Tools You Must Implement

Market Data Tools:

- `get_stock_quote` — Live/latest quote for an NSE/BSE ticker (LTP, change, volume, market cap, P/E, 52W range)
- `get_price_history` — Historical OHLCV for a ticker over a date range
- `get_index_data` — Current value and composition of Nifty 50, Sensex, sectoral indices
- `get_top_gainers_losers` — Today's top movers on NSE/BSE
- `get_technical_indicators` — SMA, EMA, RSI, MACD, Bollinger Bands for a ticker

Fundamental Analysis Tools:

- `get_financial_statements` — Income statement, balance sheet, cash flow (annual/quarterly)
- `get_key_ratios` — P/E, P/B, ROE, ROCE, debt/equity, current ratio, dividend yield
- `get_shareholding_pattern` — Promoter, FII, DII, retail holdings over time
- `get_quarterly_results` — Latest quarterly results with YoY/QoQ comparison

Mutual Fund Tools:

- `search_mutual_funds` — Search schemes by name, fund house, or category
- `get_fund_nav` — Latest and historical NAV for a scheme
- `compare_funds` — Side-by-side comparison of 2-5 mutual fund schemes

News & Sentiment Tools:

- `get_company_news` — Latest news articles for a company
- `get_news_sentiment` — Aggregated sentiment for a company over a time window
- `get_market_news` — Broad Indian market and sector news

Macro & Regulatory Tools:

- `get_rbi_rates` — Current repo rate, reverse repo, CRR, SLR, and historical changes
- `get_inflation_data` — CPI and WPI inflation time series
- `get_corporate_filings` — List recent BSE/NSE filings for a company

Cross-Source Reasoning Tools (the differentiator):

- `cross_reference_signals` — Pull from multiple sources, identify confirmations or contradictions. Example: "TCS share price fell 4% [NSE], but quarterly results show 8% revenue growth [BSE filing], and FII holding increased 2% [shareholding data]. News sentiment is negative due to US recession fears [NewsAPI], not company-specific issues."

- `generate_research_brief` — Synthesise price data, fundamentals, MF exposure, news, filings, and macro context into a structured research note with evidence citations from each source.
- `compare_companies` — Side-by-side comparison of 2-5 companies across price performance, fundamentals, shareholding, news sentiment, and mutual fund exposure.

MCP Resources

- `watchlist://{user_id}/stocks` — User's personal watchlist (auth-scoped)
- `research://{ticker}/latest` — Most recent cached research brief
- `market://overview` — Nifty, Sensex, Bank Nifty, top gainers/losers, FII/DII flows summary
- `macro://snapshot` — Latest repo rate, CPI, GDP growth, forex reserves, USD-INR

MCP Prompts

- `quick_analysis` — Fast overview: quote + key ratios + recent news for a ticker
- `deep_dive` — Comprehensive: pulls all available data, cross-references, generates full research brief
- `sector_scan` — Compare top companies in a Nifty sector index across fundamentals and sentiment
- `morning_brief` — Daily summary: market overview + watchlist updates + macro changes + key news

PS1 Tier Addendum

	Free	Premium	Analyst
`quick_analysis` prompt	✅	✅	✅
`deep_dive`, `sector_scan`, `morning_brief` prompts	❌	✅	✅
`cross_reference_signals`, `generate_research_brief`, `compare_companies`	❌	❌	✅

Must-Show Demo Scenario

1. **Free user** connects → asks "What's happening with HDFC Bank?" → gets quote + news → tries fundamentals → hits 403

2. **Premium user** runs `quick_analysis` for INFY → gets quote + ratios + news + shareholding
3. **Analyst** runs `deep_dive` for RELIANCE → system pulls from 5+ sources → generates research brief with citations from each source → saved to `research://RELIANCE/latest`

Must-Show Cross-Source Moment

`cross_reference_signals` for any ticker must combine data from at least 3 different APIs and explicitly state what confirms vs contradicts across sources.

Must-Show Auth Boundary

A free user attempts `get_financial_statements` → receives 403 with `insufficient_scope` → upgrades to Premium → same call succeeds.

🔀 [PS2] USE CASE 2: Portfolio Risk & Alert Monitor

"Your portfolio's always-on watchdog"

***Pick this if** you want to go deep on real-time state management, event-driven architecture, and personalised intelligence. You'll build something that feels alive — it watches, detects, and alerts. MCP resources and subscriptions become your primary primitives, not just tools.*

What You Build

A portfolio monitoring system. The user provides their holdings (list of stocks + quantities). The MCP server continuously monitors for risk signals — concentration, sector tilt, macro sensitivity, sentiment shifts — and surfaces alerts. Think of it as a personal risk analyst that never sleeps.

MCP Tools You Must Implement

Portfolio Management Tools:

- `add_to_portfolio` — Add a stock with quantity and avg buy price to user's portfolio
- `remove_from_portfolio` — Remove a holding
- `get_portfolio_summary` — Current value, P&L, allocation breakdown by stock and sector

- `portfolio_health_check` — Concentration risk, sector exposure, top holdings as % of total

Risk Detection Tools:

- `check_concentration_risk` — Flag if any single stock > 20% of portfolio or any sector > 40%
- `check_mf_overlap` — Given the user's stocks, check overlap with popular large-cap MF schemes. Are they unintentionally duplicating exposure?
- `check_macro_sensitivity` — Given holdings, assess which are sensitive to RBI rate changes, inflation, forex moves. Pull current macro data and flag if conditions are adverse.
- `detect_sentiment_shift` — For each holding, compare last 7-day news sentiment vs 30-day baseline. Flag significant shifts.

Market Data Tools (shared base):

- `get_stock_quote` — Live/latest quote
- `get_price_history` — Historical OHLCV
- `get_index_data` — Index values and composition
- `get_top_gainers_losers` — Today's movers

Supporting Tools:

- `get_shareholding_pattern` — Promoter, FII, DII holdings
- `get_company_news` — Latest news for a stock
- `get_news_sentiment` — Aggregated sentiment
- `get_rbi_rates` — Current macro rates
- `get_inflation_data` — CPI/WPI data
- `search_mutual_funds` / `get_fund_nav` — MF data for overlap checks

Cross-Source Reasoning Tools (the differentiator):

- `portfolio_risk_report` — Pulls market data + fundamentals + macro + news + MF overlap for entire portfolio. Produces a structured risk report: "Your portfolio is 45% IT sector [derived from holdings]. TCS and Infosys are both in your top 5 MF schemes [MFapi cross-ref]. RBI held rates steady [RBI DBIE], but USD-INR weakened 2% this month [macro], which historically pressures IT margins [news correlation]."
- `what_if_analysis` — "What happens to my portfolio if RBI cuts rates by 25bps?" Cross-references each holding's rate sensitivity with historical price reactions to past rate cuts.

MCP Resources

- `portfolio://{user_id}/holdings` — User's current portfolio (persisted, auth-scoped)
- `portfolio://{user_id}/alerts` — Active risk alerts for the portfolio
- `portfolio://{user_id}/risk_score` — Overall portfolio risk score (updated on each health check)
- `market://overview` — Market summary
- `macro://snapshot` — Latest macro indicators

Resource Subscriptions (Key Differentiator for This Use Case)

- Subscribe to `portfolio://{user_id}/alerts` — notify client when a new risk signal is detected (e.g., a holding's sentiment shifts negative, or concentration risk breaches threshold)
- Subscribe to `market://overview` — notify on significant market moves affecting portfolio holdings

MCP Prompts

- `morning_risk_brief` — Daily: portfolio value + overnight news for holdings + any new alerts + macro changes that affect holdings
- `rebalance_suggestions` — Based on current risk flags, suggest trades to reduce concentration or sector tilt
- `earnings_exposure` — Which holdings have upcoming earnings? What's the risk from each?

PS2 Tier Addendum

	Free	Premium	Analyst
`add_to_portfolio`, `remove_from_portfolio`, `get_portfolio_summary`	✅	✅	✅
`portfolio_health_check`, `check_concentration_risk`	❌	✅	✅
`check_mf_overlap`, `check_macro_sensitivity`, `detect_sentiment_shift`	❌	✅	✅
`portfolio_risk_report`, `what_if_analysis`	❌	❌	✅

`morning_risk_brief`, `rebalance_suggestions` prompts	❌	✅	✅
Resource subscriptions on `portfolio://{user_id}/alerts`	❌	✅	✅

Must-Show Demo Scenario

2. **Runs `portfolio_health_check`** — system flags concentration risk (e.g., "38% in IT sector — above 30% threshold")
3. **Runs `check_mf_overlap`** — "HDFC Bank and ICICI Bank are in 7 of the top 10 large-cap MF schemes. Your banking exposure is higher than you think."
4. **Runs `portfolio_risk_report`** — full cross-source analysis combining market data, macro, news sentiment, and MF overlap

Must-Show Cross-Source Moment

`portfolio_risk_report` must combine: current prices [NSE], sector mapping [derived], macro indicators [RBI], news sentiment [NewsAPI/Finnhub], and MF overlap [MFapi] — into a single coherent risk narrative with citations.

Must-Show Auth Boundary

- Free user can add portfolio and see basic summary → attempts `portfolio_risk_report` → 403
- Premium user can run health check and overlap → attempts `what_if_analysis` → 403
- Analyst gets everything including `what_if_analysis` and full risk reports

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🔀 [PS3] USE CASE 3: Earnings Season Command Center

"The results season war room"

***Pick this if** you love working with documents, temporal logic, and structured data extraction. You'll go deep on BSE filing PDFs, build pre-vs-post earnings workflows, and create a system that can parse a quarterly result, compare it to expectations, and explain the market's reaction — all from raw data.*

What You Build

An earnings-focused intelligence system. Built for results season (Jan-Feb, Apr-May, Jul-Aug, Oct-Nov), where 50+ major companies report quarterly. The system tracks the earnings calendar, builds pre-earnings profiles, parses post-earnings filings, and cross-references the market reaction with actual numbers.

MCP Tools You Must Implement

Earnings Calendar Tools:

- `get_earnings_calendar` — Upcoming results dates for Nifty 50 / Nifty 500 companies (next 2 weeks). Source: BSE announcements
- `get_past_results_dates` — Historical results announcement dates for a company

Pre-Earnings Analysis Tools:

- `get_eps_history` — Historical EPS (quarterly) for a company with YoY/QoQ trend
- `get_pre_earnings_profile` — For a given ticker: last 4 quarters of results + current key ratios + shareholding changes (did FIIs buy/sell before results?) + options activity (put/call OI at key strikes) + recent news sentiment
- `get_analyst_expectations` — Consensus estimates if available from Finnhub or Alpha Vantage. **Practical note:** Free-tier coverage for Indian stocks is limited. It is perfectly acceptable to use historical EPS growth rate extrapolation as the estimate, or hardcode a small set of estimates for demo purposes. Don't spend hackathon time chasing data that doesn't exist in free APIs.

Post-Earnings Analysis Tools:

- `get_filing_document` — Retrieve the actual quarterly result filing PDF from BSE
- `parse_quarterly_filing` — Extract revenue, profit, EPS, margins from the filing PDF/HTML into structured data. **Practical note:** BSE filing PDFs have inconsistent formats across companies. Teams may use LLM-based PDF extraction (send PDF content to LLM with a structured output prompt) rather than building custom parsers. The focus is on MCP integration and cross-source reasoning, not building a PDF parsing engine from scratch.
- `compare_actual_vs_expected` — Actual numbers [from filing] vs estimates [from pre-earnings]. Beat/miss/inline verdict with magnitude.
- `get_post_results_reaction` — Price change on results day and next 2 trading days [NSE price data]. Volume spike detection.

Cross-Source Reasoning Tools (the differentiator):

- `earnings_verdict` — The full picture: "Infosys reported Q3 revenue of ₹41,764 Cr (+5.2% YoY) [BSE Filing #12345], beating consensus estimate of ₹41,200 Cr [Finnhub]. EPS came in at ₹16.43 vs expected ₹15.90. However, the stock dropped 3.1% on results day [NSE]. FII holding had already decreased 1.8% in the prior quarter [shareholding data], and management commentary on deal pipeline was cautious [news sentiment]. The sell-off appears pre-positioned and guidance-driven, not results-driven."
- `earnings_season_dashboard` — Across all companies reporting this week: who beat, who missed, sector-wise trends, aggregate market reaction.
- `compare_quarterly_performance` — Side-by-side: 2-4 companies in the same sector, same quarter. Revenue growth, margin trends, market reaction, shareholding changes.

Supporting Tools (shared base):

- `get_stock_quote`, `get_price_history`, `get_index_data`
- `get_key_ratios`, `get_shareholding_pattern`
- `get_option_chain` — Full option chain with Greeks for a given stock/index and expiry (used in `get_pre_earnings_profile` for options activity)
- `get_company_news`, `get_news_sentiment`
- `get_rbi_rates` (for context on banking/NBFC earnings)
- `search_mutual_funds` / `get_fund_nav` (for MF exposure changes around results)

MCP Resources

- `earnings://calendar/upcoming` — Next 2 weeks of earnings dates
- `earnings://{ticker}/latest` — Most recent parsed quarterly result for a ticker
- `earnings://{ticker}/history` — Last 8 quarters of structured results data
- `filing://{ticker}/{filing_id}` — Parsed BSE filing content as structured resource
- `market://overview` — Market summary

MCP Prompts

- `earnings_preview` — Pre-results: historical EPS trend + estimates + shareholding shifts + options activity + sentiment. "Here's what to expect from TCS results tomorrow."
- `results_flash` — Post-results: parse filing + beat/miss verdict + market reaction + quick take. "Infosys beat on revenue, missed on margins, stock down 2%."
- `sector_earnings_recap` — "How did IT sector do this earnings season?" Aggregate across Infosys, TCS, Wipro, HCL Tech.
- `earnings_surprise_scan` — Scan recent results for biggest positive/negative surprises vs expectations.

PS3 Tier Addendum

	Free	Premium	Analyst
`get_earnings_calendar`, `get_past_results_dates`	✅	✅	✅
`get_eps_history`, `get_pre_earnings_profile`, `get_analyst_expectations`	❌	✅	✅
`get_post_results_reaction`, `compare_actual_vs_expected`	❌	✅	✅
`get_filing_document`, `parse_quarterly_filing`	❌	❌	✅
`earnings_verdict`, `earnings_season_dashboard`, `compare_quarterly_performance`	❌	❌	✅
`earnings_preview`, `results_flash` prompts	❌	✅	✅
`sector_earnings_recap`, `earnings_surprise_scan` prompts	❌	❌	✅

Must-Show Demo Scenario (e.g., "INFY") → system pulls EPS history, shareholding changes, options data, news sentiment → structured preview

2. **Post-earnings:** Run `results_flash` after a company has reported → system fetches BSE filing → parses revenue/profit/EPS → compares to estimates → checks market reaction → delivers verdict
3. **Cross-company:** Run `compare_quarterly_performance` for TCS vs INFY for the same quarter → side-by-side on all dimensions

Must-Show Cross-Source Moment

`earnings_verdict` must combine: parsed filing data [BSE PDF], price reaction [NSE], shareholding changes [BSE/NSE], news sentiment [NewsAPI/Finnhub], and estimates [Finnhub/Alpha Vantage] — into a single narrative explaining the "why" behind the market's reaction.

Must-Show Auth Boundary

- Free user asks about upcoming earnings → gets calendar → tries `get_filing_document` → 403
- Premium user gets pre-earnings profile with ratios and shareholding → tries `earnings_verdict` (cross-source) → 403
- Analyst runs full `earnings_verdict` and `sector_earnings_recap`

📊 Evaluation Criteria (Same for All Use Cases)

MCP Server Design & Compliance (25%)

- Correct tools, resources, and prompts with proper schemas
- Streamable HTTP transport, capability negotiation, tier-aware tool discovery
- Error handling, pagination, caching
- Tools return structured JSON — server provides data, client handles narrative

Authentication & Authorisation (25%)

- OAuth 2.1 + PKCE working end-to-end
- Protected Resource Metadata endpoint correctly configured
- Tiered access enforced across tools, resources, and prompts
- Rate limiting per tier with proper 429 responses
- Proper 401/403 with WWW-Authenticate for step-up auth
- Auth server separated from MCP server
- Upstream API key isolation

Cross-Source Reasoning Quality (25%)

- Do cross-source tools genuinely combine data from multiple APIs?
- Are contradictions and confirmations across sources explicitly identified?
- Are evidence citations specific (e.g., "NSE: RELIANCE LTP ₹2,456", "BSE Filing ID 12345: Q3 revenue ₹2.3L Cr")?
- Does the cross-source output add insight beyond what any individual API provides?

System Design & Technical Depth (15%)

- Caching strategy (appropriate TTLs, upstream quota awareness)
- Rate limiting (per-user, per-tier, upstream-aware)
- Token validation, error handling for upstream failures (graceful degradation)
- Audit logging

Demo & Usability (10%)

- Works with at least one MCP client (Claude Desktop, VS Code, or custom)
- Demo covers: auth flow → tool discovery → use tools → permission boundary → tier upgrade
- Clear setup instructions and API documentation

🖥️ Deployment Requirement

The MCP server must be runnable as a remote server over HTTP:

- Hosted server accessible over HTTP (localhost or cloud-deployed)
- Docker Compose setup (MCP server + auth server + cache/DB)
- README with setup instructions and list of required API keys with sign-up links

Bonus points for:

- One-command Docker Compose (MCP server + Keycloak + Redis)
- Working integration with Claude Desktop demonstrated live
- `.env.example` with clear instructions for obtaining free API keys
- Health-check endpoint showing upstream API status and remaining quotas

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📦 Expected Deliverables

1. **Working MCP Server** — Deployed and testable with at least one MCP client, with full OAuth flow and tier-based access
2. **Architecture Diagram** — MCP server components, auth flow (client → auth server → token → MCP server), upstream API integration, caching layer, tier → scope → permission mapping
3. **Technical Explanation** covering:
    - MCP primitive design decisions (what's a tool vs. resource vs. prompt and why)
    - OAuth 2.1 implementation: auth server choice, PKCE, token validation
    - Tier-based access control: scope definitions, enforcement points

    - Upstream API integration: key management, caching, rate limiting, quota management
    - Cross-source reasoning: how signals from different APIs are combined
    - Security: token audience binding, API key isolation, audit logging
4. **API Documentation** — Complete reference of all tools (input/output schemas), resources (URI patterns), prompts (with arguments), and scope requirements for each operation